MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPES 7190 AND 7191

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of specification MIL-E-1.

DESCRIPTION: Triode, hydrogen

See figures 1A and 1B

Mounting position: Any

Weight: 0.5 ounce (14.2 grams) nominal

ABSOLUTE RATINGS:

Parameter:	Ef	epy	epx	Ebb	Ecc	egx	egy	ib	Ip
Unit:	V	kv	kv	Vdc	Vdc	v	v	a	Aac
Maximum:	Note 1	1.2 (see note 2)	1.0 (see note 3)	---	---	200	400	20	1.0
Minimum:	---	---	---	300	---	---	175 (see note 4)	---	---
TEST CONDITIONS:	6.3	1.2	---	---	0	---	130	---	---

ABSOLUTE RATINGS:

Parameter:	Ib	tj	tk	$\frac{dik}{dt}$	Pb	Ehk	prr	TA	Barometric pressure, reduced
Unit:	mAdc	ns	sec	a/µs	---	V	---	°C	mmHg
Maximum:	50	5 (see note 5)	---	400	0.1×10^9	+25	Note 6	+125	---
Minimum:	---	---	30	---	---	-100	---	-60	20
TEST CONDITIONS:	---	---	30	---	---	---	5,000	---	---

GENERAL:

Qualification - Required

Ⓖ denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
---	Direct current holdoff	9	egy = 0; Rp = Rg = 1,000 ohms; tk = 30	0.65	II	Ebb	1,600	---	Vdc
---	Grid short and continuity	10	Ecc = +200 Vdc; Rg/Ic = 5 mAdc (max); epy = 0	0.65	II	---	---	---	---
1201	Short and discontinuity detection	-		0.40	II	---	---	---	---
3241	Heater current	-		0.65	II	If	1.5	2.0	A
3267	Instantaneous starting	7, 12, 19	epy = 1,000 v	0.65	II	---	---	---	---
(6) 3247	DC anode voltage for conduction	13		0.65	II	Ebb	---	100	Vdc
(6) 3246	Operation (1)	7, 14, 19	epy = 1.4 kv(min); Ib = 20a(min)	0.65	II	egy	---	130	v
3204	Voltage drop	7, 15	tp = 1.0 µs ±10%; prr = 2,000	0.65	II	etd	---	90	v
	Quality conformance inspection, part 2								
1031	Sweep-frequency vibration (1)	8, 11	t = 4 minutes; Ebb = 1,600 V; egy = 0; Rp = Rg = 1,000 ohms; Ehk = -100 V	---	---	---	---	---	---
---	Sweep-frequency vibration end points:								
3246	Operation (1)	-		---	---	egy	---	130	v
3261	Time jitter	-	epy = 500 v (max)	---	---	tj	---	4.0	ns
1041	Shock	-	Ef = 6.3 V; no voltages applied; 720 G at 1 ms	---	---	---	---	---	---
---	Shock end points:								
3246	Operation (1)	7		---	---	egy	---	130	v
3261	Time jitter	7	epy = 500 v(max)	---	---	tj	---	5.0	ns
3241	Heater current	-	Ef = 7.5 V	---	---	If	1.6	2.2	A
1336	Heater-cathode leakage	-	Ehk = -100 Vdc Ehk = +50 Vdc	---	---	Ihk	---	100	µAdc
3256	Anode delay time	-	Operation (1); t = 120	---	---	tad	---	0.6	µs
3256	Anode delay time drift	16	Anode delay time	---	---	Δtad	---	0.15	µs
3261	Time jitter	7	epy = 500 v(max); egy = 175 v	---	---	tj	---	4.0	ns

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3								
---	Life test	7, 17	Group C; e_{gy} = 1,250 v; e_{gy} = 140 v; I_b = 20a; prr = adjust for I_p = 1.0 Aac; t = 500 hours	---	---	---	---	---	---
---	Group C, establishment of prerelease eligibility	-		---	---	---	---	---	---
---	Life - test end points:								
3246	Operation (1)	-		---	---	e_{gy}	---	140	v
3256	Anode delay time drift	-	e_{gy} = 140 v	---	---	Δt_{ad}	---	0.2	µs
3261	Time jitter	-	e_{gy} = 175 v	---	---	t_j	---	5.0	ns
3204	Voltage drop	-		---	---	e_{fd}	---	130	v
1506	Heater-cycling life	18	Ef = 7.0 V; Ehk = -100 Vdc; Ec = Ebb = 0	---	10 tubes	---	---	---	---
---	Heater-cycling life-test end points:	-							
1336	Heater-cathode leakage	-	Ehk = +50 Vdc Ehk = -100 Vdc	---	---	I_{hk}	---	125	µA
1002	Barometric pressure, reduced (type 7191)	20	Pressure = 20±5 mmHg; Ebb = 1,400 V; e_{gy} = 0; R_p = R_g = 3,000 ohms; Ehk = -100 V	---	---	---	---	---	---
1031	Sweep-frequency vibration (2)	7, 8, 20	t = 120 minutes in 3 planes; 40 min/plane	---	---	---	---	---	---
1031	Vibration fatigue	20	No voltages applied	---	---	---	---	---	---
3246	Operation at elevated ambient temperature	7, 20	TA = +125°C; t = 300 (min)	---	---	e_{gy}	---	130	v

NOTES:

1. It is recommended that a heater voltage, as determined from figure 2, be utilized whenever I_p is less than 0.75 Aac. The heater voltage must be held within ±5 percent of the values shown on figure 2.
2. For cold-starting applications, where anode voltage is applied instantaneously, the power-supply filter design shall be such that the maximum permissible e_{gy} is 1,000 v and cannot be attained in less than 0.04 second. It is recommended that the driver pulse be applied prior to the application of anode voltage.
3. In pulsed operation, the peak inverse voltage shall not exceed 500 v during the first 25 µs after the pulse.
4. The value shown is valid only for the following pulse shape: time of rise = 0.5 µs maximum, grid pulse duration = 2.0 µs minimum. Impedance of drive circuit = 1,000-ohms maximum.

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NOTES: - Continued

5. The rated time jitter of 5 nanoseconds (ns) is applicable only at peak forward anode voltages in excess of 500 v epy, and with a driver pulse in excess of 250 v egy. At values of epy less than 500 v or egy less than 250 v, the time jitter maximum is 10 ns.

6. Under reduced ratings, operating frequencies up to 50 kHz are feasible.

7. The circuit constants shall be chosen so that at epy = 1.2 kv under resonant charging conditions: dik/dt = 400a/µs minimum, ib = 20a, tp = 0.50 µs ±10 percent, prr = 5,000, and Rgl = 10,000 to 30,000 ohms. The grid pulse characteristics shall be tp = 2.0 µs maximum, tr = 0.5 µs, and driver impedance = 1,000-ohms minimum.

8. Heater-cathode shorts of 100,000 ohms or less, having a duration of 100 µs or more, are cause for rejection.

9. Inability to hold off dc anode voltage, when tapped with a cork hammer using 3-inch strokes constitutes a failure.

10. Monitor grid-cathode voltage to indicate short or open when tube is tapped with a cork hammer using 3-inch strokes. Indication of short or open is cause for rejection.

11. Failure to holdoff dc anode voltage during the vibration period shall constitute a failure.

12. The tube shall operate satisfactorily on pushbutton starting within three attempts when anode voltage is applied in such a manner as to rise from 0 to 1,000 v within 0.03 second (the filter design shall be such that at least 500 v is reached within 0.015 second). Any tube failing to start within three attempts shall be considered a failure.

13. No voltages shall be applied to the grid or anode of the tube under test for at least 10 minutes prior to performing this test.

14. The tube shall operate continuously for 5 minutes without evidence of continuous conduction, anode heating, or loss of control.

15. Using a calibrated oscilloscope, adjust epy for a peak cathode current of 20 amperes. Measure the voltage drop between anode and cathode from the zero reference line to the center of the tube drop porch (see figure 3).

16. During the interval between 2 minutes and 7 minutes of the tad test, the anode delay time drift relative to the tad value observed on the anode delay time test, shall not exceed the value specified herein.

17. Life test shall be operated with the tube in a horizontal position. Operation shall consist of 12 minutes of operation (heater, trigger, and anode voltage) and 12 minutes off. The application of anode voltage may be delayed a maximum of 60 seconds after the application of heater voltage. Only anode operating time shall be considered test time. The average life of the sample shall be not less than 90 percent of the duration of the test.

18. One defect is allowed if the sample from the preceding lot contained zero defects. Otherwise, no defects are allowed.

19. This test is to be the first test performed at the conclusion of the holding period.

(G) 20. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

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Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

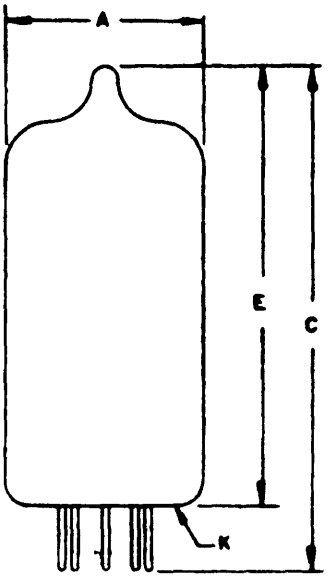
User activities:
Army - MU
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

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PIN CONNECTIONS	
1	Grid
2	Cathode
3	Heater
4	Heater
5	Cathode
6	Anode
7	Cathode

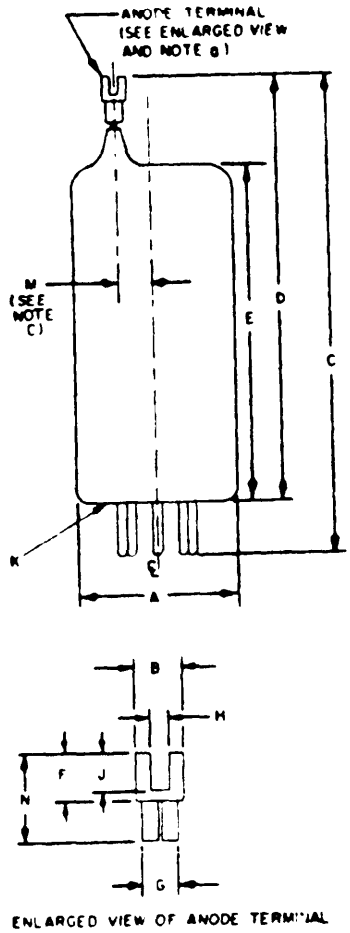
Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	.804 (20.42)	.844 (21.44)
C		2.125 (53.98)
E		1.875 (47.63)
Quality conformance inspection, part 3 (SEE NOTE d)		
K	Base: E7-1	

NOTES:

a. Anode terminal material shall be nickel.
b. Pins must be gold plated, if made of tungsten.
c. This dimension lies in a plane passing through the centerline of the tube and pin 6.
d. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1A. Outline drawing of electron tube type 7198.

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ENLARGED VIEW OF ANODE TERMINAL

PIN CONNECTIONS	
1	Grid
2	Cathode
3	Heater
4	Heater
5	Cathode
6	No connection (do not use No. 6 pin socket terminal as tie point)
7	Cathode

Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Quality conformance inspection, part 2		
A	.804 (20.42)	.844 (21.44)
C		2.478 (62.94)
D		2.228 (56.54)
Quality conformance inspection, part 3 (SEE NOTE d)		
B	.115 (2.92)	.135 (3.43)
E		1.593 (40.46)
F	.090 (2.29)	.110 (2.79)
G	.087 (2.21)	.101 (2.57)
H	.037 (.94)	.047 (1.19)
J	.070 (1.78)	.090 (2.29)
K	Base: E7-1	
M	.075 (1.91)	.175 (4.45)
N	.210 (5.33)	.230 (5.84)

NOTES:
a. Anode terminal material shall be nickel.
b. Pins must be gold plated, if made of tungsten.
c. This dimension lies in a plane passing through the centerline of the tube and pin 6.
d. Dimensions shall be checked during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

FIGURE 1B. Outline drawing of electron tube type 7191

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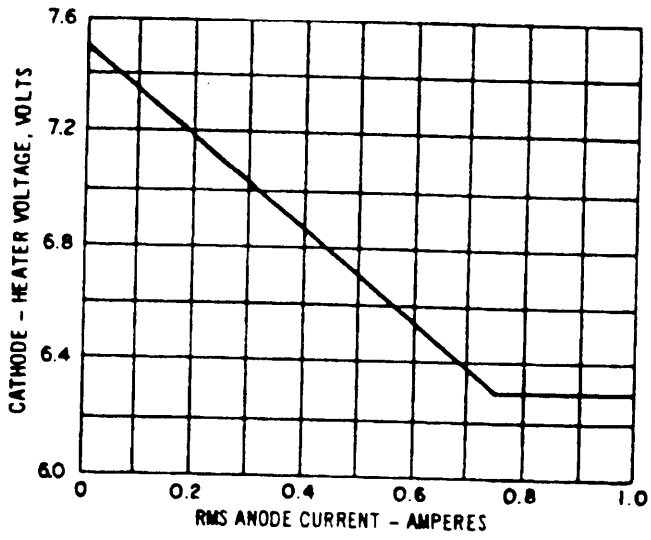


FIGURE 2. Recommended cathode heater voltage, electron tube types 7190 and 7191.

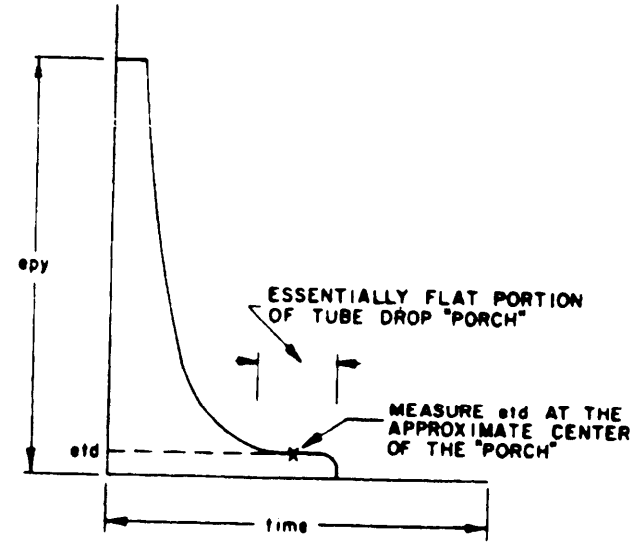


FIGURE 3. Tube drop porch.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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